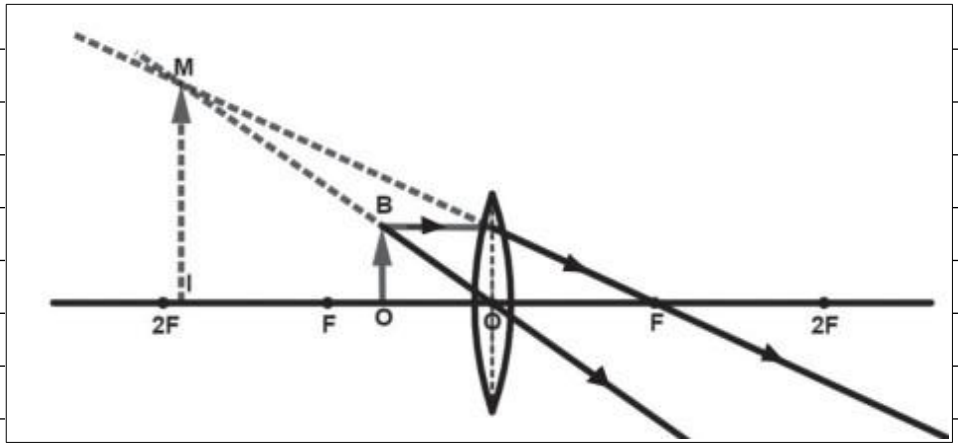
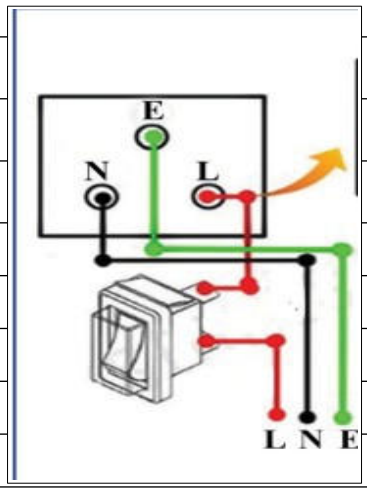
SNO	SSLC EXAMINATION-MARCH-2026 ANSWERKEY---PHYSICS	MARKS
	Section-A	
1	**(C)---Joule heating**	1
2	**(c) The Statement is correct but the Reason is incorrect.**	1
3	**(d) (ii) and (iv) are correct.**	1
4	**(c) ABF and DEC**	1
	Section-B	
5	**a)minimum distance is 17m**	1
	b)d_1 =vx t_1/2 =340x0.2/2=34m	
	D_2 =vx t_2/2 =340x 0.3/2 =51m =34+51=85m	1
6		2
7.A	**Tyndall Effect---Dissolve the sodium thiosulfate in the water. Then, carefully add the sulfuric acid and stir.** **the phenomena described—the blue sky and red sunsets—are caused by Rayleigh scattering. This occurs when sunlight interacts with small particles and gas molecules in the Earth's atmosphere.**	1 1
B	**a)tyndall effect-colloidal medium**	1
	b)size of the particle and wavelength	1
8	**a)"X" represents the Main Switch**	1
	b) 	1

9	**a)MA=50/10=5**	**1**
	b)Increase the radius of the wheel:Decrease the radius of the axle	**1**
10	**a)electrical energy into heat energy**	**1**
	b)P=E/t =7200000j/7200s =1000w=1kW	**1**
	OR	
B	**a)electrical energy to mechanical energy**	**1**
	b)W=Pxt =400x60 =24000j	**1**
11	**Fleming's Left-Hand Rule.--Hold the thumb, first finger, and second finger of your left hand perpendicular to each other. If the First finger points in the direction of the magnetic field and the seCond finger in the direction of the electric current, then the thuMb will indicate the direction of the force experienced by the conductor.**	**1** **1**
	Section-c	
12	**a)A is Green D is Blue**	**1**
	b)Red, Green (A), and Blue (D).	**1**
	c)Red and Cyan (C).	**1**
13	**a)nichrome**	**1**
	b)H=V^2 *Rxt 40000/* 60x300 =200000j	**1**
	c)The amount of heat produced will increase.Since the voltage and time remain constant, the heat produced is inversely proportional to the resistance	**1**
14 A	**a)(i) the tweezers/forceps**	**1**
	b)It states that for a lever in equilibrium the clockwise moment must be equal to the anticlockwise moment.	**1**
	c)MA= effort arm/load arm 2 =1/load arm load arm=0.5m	**1**
B	OR	
	a)A lever is a simple machine consisting of a rigid bar or beam that is capable of rotating about a fixed point called the fulcrum.	**1**
	b)	**1**

Lever Type	Component in the Middle
Second Order Lever	The **Load** (L) is in the middle.
Third Order Lever	The **Effort** (E) is in the middle.

	c)W_1 x d_1 =W_2 x d_2 60x 25 =W_2 x30	
	W_2 =50 kg wt	**1**
15	**a)A solenoid is a coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder**	**1**
	b)No, the statement is incorrect.	**1**
	c)Increasing the amount of current,Increasing the number of turns per unit length	**1**
16	**a)longitudinal wave.**	**1**
	b)C (Compression) R (Rarefaction)	**1**
	c)v=fλ f=350/0.01 =35000Hz	
	The calculated frequency is 35,000 Hz. Since the human audible range is between 20 Hz and 20,000 Hz, this wave is not audible to humans because it falls into the ultrasonic range (above 20,000 Hz).	**1**
17 A	**a)1/v +1/u =1/f** **1/v= -2/20 -1/20** **1/v= -3/20** **v=-20/3=-6.67cm**	**2**
	b)Nature: The image is virtual (since is negative). **Orientation: The image is erect (upright).** **Size: The image is diminished**	**1**
B	**OR**	
	a)1/f =1/v -1/u **1/15 =1/-7.5-1/15** **1/u= -1/5** **u= -5cm The distance between the lens and the object is 5cm**	**2**
	b)Magnifying Glass ,Simple Microscope	**1**
	Section-D	
18	**a)The bulb in the second circuit glows only at the moments of turning the switch ON or OFF because an induced electromotive force (EMF) is created only when there is a change in magnetic flux through the second coil.**	**1**
	b)To make the bulb glow continuously, the DC source (battery) in the first circuit should be replaced with an AC.Mutual Induction.Consider two coils kept close to each other. When the intensity or direction of current in one of them changes, the magnetic field around it changes. As a	**2**

	result, an emf is induced in the second coil.	
	c)V_s /V_p =N_s /N_p **400/V_p =200/100** **V_p =200v**	**1**
B	**OR**	
	a)Transformer.It is used to increase (step-up) or decrease (step-down) the alternating current (AC) voltage	**1**
	b)V_p xI_p =V_s x I_s **20 x I_p =100x 1** **I_p =100/20 =5A**	**1**
	c)The secondary coil has more turns than the primary coil	
	The primary coil wire is thicker than the secondary coil wire.	**2**